# Pre-registered protocol: a global sensitivity audit of the scenario-generation pipeline of Li et al. (2025), *Nature Climate Change*

Modelling of the Modelling Process (MoMP) series — optimal-transport GSA case

Version 0.1 — [TBD: date of Zenodo deposit] — DOI: [TBD: minted at deposit]

**Abstract**

We pre-register a global sensitivity analysis (GSA) of the scenario-generation pipeline of Li, Zhu, McJeon, Byers, Zhou & Ou (2025), "Using deep learning to generate key variables in global mitigation scenarios", *Nat. Clim. Change* 15:760–768 (doi:10.1038/s41558-025-02352-8). The object of analysis is not the climate or the AR6 ensemble but the pipeline itself: the analyst's methodological choices are the input factors, the generated multivariate ensemble and the paper's validation claims are the outputs, and sensitivity is measured by optimal-transport (Wasserstein) displacement of the synthetic ensemble, estimated from given data. This protocol freezes, before any GSA run: the operational rules for the paper's claims (§4), the factor space (§5), the experimental design and compute budget (§6), the estimators and inference procedure (§7), and the decision criteria (§8). The completed replication record (Phase 0, §3) is reported for transparency. At the time of this deposit, no claim rule (X1a–X5) has been evaluated at any factor setting, including the paper's own values, no retraining has been performed, and no noise-floor replicate has been run. The only pipeline stage executed is the deterministic regeneration check of Phase 0B, which evaluates no claim.

## 1 Purpose and scope

Continuing the MoMP programme, this study asks whether the properties Li et al. attribute to their synthetic-scenario framework are properties of the framework or of the specific, partly undocumented methodological choices through which it was instantiated. We audit the epistemic apparatus, not the climate–economy content: factors are analyst choices, outputs are (i) the synthetic ensemble as a multivariate distribution and (ii) pass/fail evaluations of the paper's claims under fixed operational rules.

## 2 Audited artefacts

All artefacts are pinned by version and SHA-256 in `DATA.md` of the project repository (archived with this deposit). Principal items:

- **A1** Deep-IAM code, v1.0.0, doi:10.5281/zenodo.15280653 — the version cited by the paper, the audited object.
- **A2** Deep-IAM v1.0.1 (2026-04-25). Admitted as documented supplement only for its two added files: `Secondary-data-generate.ipynb` (the generator of the deposited synthetic data; absent from v1.0.0) and `Configuration of Environments.txt` (environment pins: Python 3.8.20, TensorFlow 2.13.0, tsgm 0.0.7, scikit-learn 1.3.2, XGBoost 2.1.1). Full diff archived.
- **A3** Synthetic-scenario deposit, doi:10.5281/zenodo.15240553 (Gen/Real CSVs; validated in Phase 0).

- **B1–B2** AR6 Scenarios Database v1.1 (metadata + World timeseries), doi:10.5281/zenodo.7197970.

## 3 Phase 0 — completed replication record

Completed before this deposit; machine-readable reports archived under `results/`. Summary:

1. **Metadata fidelity.** AR6 v1.1 vetted-with-climate population reproduces the paper's Extended Data Table 1 exactly: $n = 1202$; per-category counts match digit for digit (`phase0_metadata_check.json`).
2. **First undeclared filter reconstructed (D2).** The training population (SI S3.1: 680/308/172, $n = 1160$) is reproduced exactly by: complete 2020–2100 coverage on the five feature variables (`Primary Energy|Coal/Oil/Gas`, `Final Energy|Liquids`, `Carbon Sequestration|CCS`); Kyoto Gases exempt. This rule appears nowhere in the paper or SI.
3. **Second undeclared filter reconstructed (D9).** The population underlying the paper's figures and label-transfer benchmark (Fig. 4a, ED Fig. 2, Table 1: 554/237/110, $n = 901$) is reproduced exactly by: complete 2020–2100 coverage on the five feature variables and all ten electricity fuel-level variables, Kyoto Gases exempt, raw (pre-imputation) CS availability required. Across the two silent drops the smallest category (C78) loses 43% of its vetted scenarios (193→110).
4. **Deposit validated.** After harmonizing a Mt-vs-Gt unit inconsistency (deposit in Mt, the paper's Source Data Fig. 3 in Gt), all 1,620 quantile comparisons (6 variables × 6 series × 9 years × 5 quantiles) pass at 2% relative tolerance; Gen files hold 10,000 scenarios per category.
5. **Version provenance (D7).** v1.0.0 cannot reproduce the deposited synthetic data or Fig. 5. The generating notebook appeared in v1.0.1 ten months after publication, without changelog.
6. **Non-reproducibility of training (D1).** The authors state that TSGM training is not reproducible even under fixed seeds and distribute trained weights, this motivates the noise-floor design of §6.
7. **Phase 0B — executability and determinism of the generation stage (completed).** Evaluating no claim rule, we executed the pipeline's one nominally deterministic link: the authors' shipped decoder weights (`vae_conv5`, 9 steps × 16 features, latent dim. 10) were loaded, the `TSFeatureWiseScaler` refit on the deposited Real CSVs (which the D9 reconstruction identifies as the training tensors), the latent noise drawn under the documented global seed 3, decoded, and compared to the deposited Gen CSVs (environment: Python 3.10.21, TensorFlow 2.13.0, tsgm 0.0.7, macOS/arm64; the authors used Windows x86-64, Python 3.8). Results (`results/phase0b_regeneration.json`): (i) *executability confirmed* — weights load, architectures match, generation runs end-to-end from the archived artefacts alone; (ii) *sample-level reproduction fails* under all three latent-stream readings of the documentation (TensorFlow global-seed, a second-execution probe, and the SI's NumPy reading): median elementwise relative deviation 0.33–0.45, i.e. entirely different draws; (iii) *distributional agreement in the bulk*: median quantile deviation 1.8% of the reference spread (Q90–Q10), 95th percentile 10.3%, yet ≈ 4.4× above the Monte-Carlo floor obtained from two independent 10,000-draw ensembles through the same decoder, with the excess concentrated in distribution tails and tight-spread cells (e.g. carbon sequestration in 2020); (iv) a scaler-bounds diagnostic flags no feature, consistent with the Real CSVs being the scaler-fit data. The residual is most plausibly a checkpoint mismatch (the shipped weights differing from those that generated the deposit) consistent with the repository README's account of multiple training sessions with "comparable" results. Conclusion: the generation stage is executable and distributionally faithful, but the deposited outputs cannot be tied to the documented seed or, apparently, to the shipped weights; the determinism gap the authors acknowledge for training (D1) extends to generation (catalogued as D11). This check answers no question posed in §4–§8.

The full inconsistency catalogue (D1–D10) is archived as `DISCREPANCIES.md`.

## 4 Claims and operational rules

Each claim of the target paper is translated into a rule verifiable on the pipeline's output. Populations are declared explicitly, given D2/D9. Thresholds marked [TBD: ] are to be fixed at freeze. No rule will be altered after deposit except under the deviations policy (§9).

**X1a (distributional match).**
"[The VAE] closely matches the original AR6 scenarios in terms of mean and the spread, particularly for C1234." *Rule:* for each of the six key variables and each category, the median over years of the absolute relative deviation between synthetic and real quantiles (Q10, Q25, Q50, Q75, Q90; real population = the 901-set) is $\leq \tau_1$, with $\tau_1$ = [TBD: propose 0.15]; "particularly C1234" holds if the C1234 deviation is the smallest of the three categories.

**X1b (narrower distributions).**
"Synthetic scenarios have narrower distributions for all climate categories." *Rule:* $\mathrm{IQR}_{\mathrm{syn}} < \mathrm{IQR}_{\mathrm{real}}$ for all six variables and all three categories at 2050 and 2100.

**X2 (correlation replication).**
"The synthetic scenarios also retained the correlations between key variables . . . matching closely those observed in the original AR6 scenarios." *Rule:* (i) the maximum absolute elementwise difference between synthetic and real correlation matrices (six variables; 2030, 2050, 2100) is $\leq \tau_2$ = [TBD: propose 0.20]; (ii) the paper's reported temporal ordering (coal–Kyoto correlation stronger in 2030 than 2100) holds in the synthetic data.

**X3 (internal consistency).**
"Among the 30,000 synthetic scenarios, 258 exhibited lower sum-check errors than the 1.2% average error in the AR6 Scenarios Database." *Rule:* the number of synthetic scenarios with electricity sum-check error $< 1.2\%$ is $\geq 258$.

**X4 (VAE superiority).**
"VAE achieved higher category label transferring accuracy . . . compared to RCGAN and CGAN" (Experiment I, synthetic→real, test population = the 901-set). *Rule:* accuracy(VAE) > accuracy(RCGAN) and accuracy(VAE) > accuracy(CGAN) in the same run.

**X5 (bias inheritance; secondary).**
The paper positions synthetic generation as mitigating "model-determined biases" and "non-random submission". *Rule:* train a source-model classifier (nine model groups, Dekker-style merge) on the real 901-set; apply to synthetic scenarios; X5 (inheritance) holds if the total-variation distance between the predicted model mix of the synthetic ensemble and the real ensemble's model shares is smaller than the distance to the uniform mix, per category. A baseline model-identifiability check on real data (held-out accuracy above the majority-class rate) gates the test.

## 5 Factor space

Factors, paper values, and perturbation ranges are frozen in `config/paper_values.yaml` (archived with this deposit); the table below summarizes. "Expensive" factors require retraining; "cheap" factors vary per trained model.

| Factor | Paper value (provenance) | Range/levels | Cost |
|---|---|---|---|
| G1 population | 1160-rule (D2; undeclared) | {1202; 1160; 901; filter variants} | exp. |
| G2 aggregation | C1234/C56/C78 (Methods) | {3-way; 4-way; 8-way; k-means labels} | exp. |
| G3 variable sel. | 5 features, RF+SHAP (SI S2) | completeness thr. [TBD: 0.6–0.9]; $n$ feat. [TBD: 4–8] | exp. |
| G4 preprocessing | mean-imputation yr×cat.; (0,1) scaling | imputation variants; scaling variants | exp. |
| G5 architecture | VAE latent dim 10; ED Table 3 | latent [TBD: 4–20]; epochs ×[TBD: 0.5–2] | exp. |
| G6 seed | seed 3; training non-reproducible (D1) | noise-floor replicates | exp. |
| G7 sample size | 10,000/category | [TBD: 1,000–30,000] | cheap |
| G8 transfer classifier | RF (code) | {RF paper; RF alt; logistic; XGBoost} | cheap |
| G9 validation metrics | Fig. 3–5 statistics | OT spec (§7); thresholds | cheap |

The D4 documentation conflict (CGAN/RCGAN epochs: SI vs ED Table 3) is handled as an explicit two-level factor nested in G5 unless adjudicated from the shipped weights before freeze.

## 6 Design and compute budget

1. **Noise floor.** $R_0$ = [TBD: propose 20] seed-only replications at paper values for all factors. The floor $\delta_0$ is the [TBD: 95th percentile] of pairwise OT distances between replicate ensembles, per category.
2. **Outer design.** Sobol' quasi-random sample of size $N_{\text{outer}}$ = [TBD: propose 512] over the expensive factors (G1–G6), one training per point, environment pinned to A2's configuration.
3. **Inner design.** Per trained model, a full or fractional grid over cheap factors (G7–G9), $N_{\text{inner}}$ = [TBD: propose 8].
4. **Stopping rule.** The budget is $N_{\text{outer}}$×(training cost); if bootstrap CIs on the leading OT indices have half-width $<$ [TBD: 0.05] at $N_{\text{outer}}/2$, the design may stop early; it may not be extended beyond $N_{\text{outer}}$ without a documented deviation.
5. **Outputs stored per run:** the generated ensemble (or sufficient statistics + per-marginal empirical quantiles), all claim evaluations X1a–X5, wall-clock, and the full factor vector.

## 7 Estimators

- **Ensemble-level sensitivity:** given-data optimal-transport sensitivity indices (Borgonovo–Chiani–Plischke class) on the synthetic ensemble as multivariate output. Distance: exact 1-D Wasserstein on each variable–year marginal, aggregated by [TBD: mean], plus sliced Wasserstein ($L$ = [TBD: 50] projections) on the joint; Sinkhorn ($\varepsilon$ = [TBD: ]) as robustness check. Partitioning per the given-data estimator with [TBD: $M$] bins; discrete factors use exact level partitions (partition-collapse guard as in our prior OT-GSA implementation).
- **Claim-level sensitivity:** for each rule X1a–X5, fraction of runs in which the claim holds, plus Sobol'-style total-order indices on the binary outcome, mirroring the companion analysis of Li P.-H. et al. (2023) for cross-paper comparability.
- **Inference:** bootstrap with $R = 1000$ resamples; 95% CIs on all reported indices.

## 8 Decision criteria

- A factor is *influential* for the ensemble output if the lower 95% CI bound of its total OT index exceeds the noise floor $\delta_0$-equivalent index.
- A claim is *robust* if it holds in $\geq$ [TBD: propose 80%] of runs across the design; *fragile* if in $< 50\%$.
- The study's headline question — are the paper's findings properties of the framework or of the analyst's choices? — is answered by the joint pattern of (i) claim fragility and (ii) the identity of the dominant factors (declared vs undeclared choices).

## 9 Deviations policy

Any departure from this protocol is logged in `DEVIATIONS.md` with date, reason, and the analysis state at the time of the change. Results affected by deviations are reported separately from the pre-registered analysis. Adjudications of [TBD: ] placeholders completed *before* the Zenodo freeze are part of this protocol. The deposited PDF contains no placeholders.

## 10 Archive contents at deposit

Protocol PDF, `DATA.md`, `DISCREPANCIES.md`, `config/paper_values.yaml` (frozen), Phase-0 reports (`results/*.json`, including the Phase 0B regeneration report), v1.0.0/v1.0.1 diff, commit hash of the analysis harness. GSA results: none, no claim rule evaluated at any factor setting (the paper's own values included), no retraining performed, no noise-floor replicate run.